Towards Available, Open Source, and Reproducible Human Activity Recognition (HAR): A Systematic Review of Wrist-Accelerometer-Only HAR Literature

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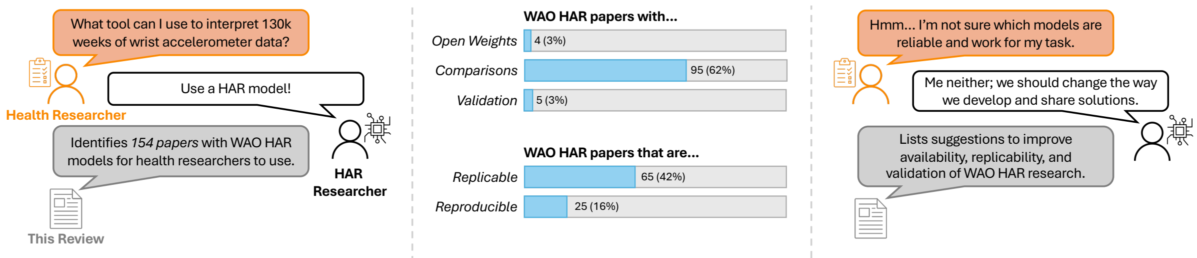

Fig. 1. The motivations (left), results (center), and implications (right) of our review. Human activity recognition (HAR) researchers work to develop models that others can use to interpret or analyze their data. Oftentimes, HAR researchers focus on novelty while health researchers may instead value availability, reproducibility, and validation of the resulting tools.

Reliable wrist-accelerometer-only (WAO) human activity recognition (HAR) models would enable improved analysis of important population-wide datasets that have already been collected. To compile a repository of such WAO HAR models, we conducted a systematic review of five databases to identify existing peer-reviewed WAO HAR literature and assessed their availability (i.e., is the work open source or open weight?), replicability, and/or reproducibility. We screened 7,822 articles, ultimately analyzing 154. We found (1) 21 publications made a WAO HAR algorithm publicly available, but *only four articles released open-weight models*; (2) the identified WAO HAR algorithms are not directly comparable because of differences in evaluation strategies and research problems investigated (e.g., real-time or generalized HAR); and (3) there is a lack of standardization in communicating work that makes assessing algorithm performance, availability, and reproducibility difficult. Ultimately, we are unable to identify "the most promising" WAO HAR model with which to analyze existing population-wide datasets. To ensure future researchers can, we propose a detailed checklist and discussion on how we ***must improve how we report, validate, and share our algorithms***.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile devices**; • **General and reference** → **Surveys and overviews**.

Additional Key Words and Phrases: Human activity, Activity recognition, Accelerometer, Wrist, Review, Reproducible research.

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1 Introduction

Health researchers rely on accurate depictions of individuals' daily activities (e.g., their physical activities, sedentary behaviors, or commutes) to establish population-wide behavioral health trends [1]. Developing "objective" human activity recognition (HAR) algorithms that can identify an activity from passively collected sensor data (e.g., smart watch or phone data) is important for behavioral health research because these automatic methods are scalable, less obtrusive than answering frequent surveys, and do not require participant recall (the current standard for obtaining activity labels [82, 83, 88, 174]). Thus, accurate, wearable-sensor-based HAR systems could be used to monitor individual's behavioral health using data gathered in a convenient manner while individuals go about their daily lives.

Motivated by the goal of measuring population-wide physical activity trends using wearable sensors, *health researchers have already collected **unlabeled**, week-long, free-living, wrist-worn accelerometry datasets from over one hundred and thirty thousand individuals through three nationally representative health surveys*: the China Kadoorie BioBank study [41], the National Health and Nutrition Examination Survey (NHANES) [2], and the UK BioBank study [61]. These data were collected from participants who wore research-grade activity monitors on the wrist because individuals increasingly wear wrist-worn devices such as smartwatches [6]. These sensors only collected accelerometer data because accelerometers consume less power than other sensors, such as gyroscopes, ensuring the devices sampled continuously for a full week on a single charge allowing week-long, 24/7 sensor measurement. In these nationwide health surveys, participants did not log their daily physical activities. The researchers who collected these data assumed that HAR algorithms would be developed to process the unlabeled, raw accelerometer data and convert these data into either overall, underlying physical activity trends or specific types of PAs, sedentary behaviors, and sleep behaviors.

Because HAR algorithms using wearable accelerometers have been developed and refined over many years (e.g., [24, 158, 197]), resulting in a large body of published, peer-reviewed literature, health researchers should have access to a variety of trained HAR models to analyze the wrist-worn accelerometer datasets being collected via public health studies. To identify the HAR models that could readily be applied to health surveillance study data, as well as other accelerometer data collected from smartwatches, we systematically reviewed the HAR literature, focusing on HAR algorithms that use a wrist-worn accelerometer. We followed the reporting guidelines in the PRISMA 2020 systematic review checklist [150][1] and investigated the following three research questions:

- **RQ1** (Development and comparability): Within the context of health-related wrist-accelerometer-only (WAO) HAR, what data are algorithms trained on? How are the resulting models evaluated and compared against existing solutions?
- **RQ2** (Availability): What are the open-weight, ready-to-use WAO HAR models? How easy are these models to use? And, how does the performance of these models differ?
- **RQ3** (Reproducibility[2]): Is work in the WAO HAR field reproducible? If not, is it replicable? What factors prevent existing WAO HAR research from being both replicable and reproducible?

By investigating these research questions, we make the following contributions:

- Categorized the 154 identified articles into one of six main contribution areas. To evaluate their specific research questions, the authors of 82 articles (53%) collected a novel activity dataset (few of which were released to the public) which may prevent reproduction of their work and prohibits others from using the dataset to answer additional questions; the authors that used public data used 27 different, existing, public datasets containing WAO HAR data. When evaluating their WAO HAR algorithms, 113 articles (73%) evaluated their WAO HAR algorithm on a single dataset, but the authors of only five articles (3%) validated

[1]The PRISMA 2020 checklist can be found at www.prisma-statement.org/prisma-2020-checklist. We have included the completed checklist for our review in our supplemental material.

[2]We define *replication* and *reproduction* as we use these terms in our review in Sec. 2.1

the performance of their model on an additional dataset. Additionally, only 95 articles (62%) compared their WAO HAR algorithm against existing baselines, and only 90 articles (59%) include class-wise results. *The lack of consensus across tasks and evaluation strategies means **it is difficult to compare or choose the best WAO HAR algorithms for health-related tasks**.*
- Identified four articles (3%) in which researchers release open-weight WAO HAR models, and a total of 21 articles (14%) in which researchers make the entire algorithms they describe available to the public. Additionally, we determined all open-weight models are usable; however, no author evaluated an open-weight model on an additional dataset and their performances may not generalize.
- Graded 65 articles (42%) as being `likely replicable` and a further 25 articles (16%) as `likely reproducible`. Because we found a minority of the described approaches to be replicable and reproducible, we discuss the *need for standardizing evaluation, making data public, and releasing open-source algorithms and open-weight models* to allow the HAR community to collaborate and converge on the best algorithmic solutions and develop analytical tools for behavioral health researchers studying physical activity.

2 Background: Definitions

We define the frequently used terms in our review in the context of the five basic steps in the human activity recognition (HAR) development cycle (or "activity recognition chain"): data acquisition, data preprocessing, data segmentation, feature extraction, and training and classification [33, 59, 153].

2.1 Definitions

These definitions apply common definitions in the machine learning and scientific communities to the context of human activity recognition.

Definition 2.1 (HAR Algorithm). A *machine learning algorithm* is a set of step-by-step instructions to train a machine learning model to make autonomous predictions on novel data [5, 81]. A ***HAR algorithm*** consists of data segmentation, preprocessing, feature extraction, and training protocols, as well as the machine learning algorithm architecture (e.g., random forest) and parameters (e.g., number of trees and CART algorithm). HAR algorithms are developed to satisfy different challenges such as performing well across a wide population (i.e., a *generalized* HAR model) or performing well on very specific activities.

Definition 2.2 (HAR Model). A *machine learning model* is a program that makes a prediction on an input without additional intervention [5, 81]. A machine learning model is obtained by training a machine learning algorithm on a specific set of training data. Thus, a ***HAR model*** is a program that takes activity data as input and outputs an activity name. In other words, a HAR model is a set of weights and a machine learning algorithm architecture obtained by training a HAR algorithm on a particular HAR dataset.

The terms *algorithm* and *model* are oftentimes used interchangeably [5], but are technically different and important to distinguish.

Definition 2.3 (Available, Open weight, and Open source). A HAR algorithm is ***available*** if the original authors make all code (i.e., the implemented algorithm) associated with their HAR algorithm public. A HAR model is ***open weight*** if the original authors make the final, trained weights and underlying machine learning algorithm architecture public [4]. A HAR solution is ***open source*** if the underlying data and algorithm are made available and the HAR model is open weight [3].

Definition 2.4 (Replicable). Scientific work is replicable if results consistent with the original work can be achieved by using the same methods and novel data [131] (i.e., someone can use the same methods and get similar results with different data). In the context of human activity recognition, we consider work to be ***replicable*** if a

HAR model can be trained using the original HAR algorithm and novel data (i.e., the procedures defining the HAR algorithm are detailed enough to be used without additional assumptions).

Importantly, this resulting HAR model may have different weights than the weights of the author's original model and thus may perform slightly differently; this model is not necessarily a direct replica of the author's original work. HAR work is replicable if 1) all the HAR algorithms are available or 2) the authors of the original work define clearly, within their publication, each step of each algorithm in their work so that an independent researcher can re-implement their algorithm without making any assumptions.[3]

Definition 2.5 (Reproducible). Scientific work is reproducible if, using the same data and processes, results computationally consistent with the original work can be achieved (i.e., the exact same results can be obtained by following the original protocols with the original data) [131]. In the context of HAR, work is ***reproducible*** if a HAR model trained using the original HAR algorithm and original data makes predictions computationally consistent with the HAR model described in the original work.

Reproducible work in HAR is necessarily replicable, uses only public data, explicitly details train/test splits (e.g., if a single train-test split is used, the exact data in the training set is specified), and, when applicable, provides the model seeds.[4]

Work in the field of HAR can be replicable without being reproducible; however, HAR work cannot be reproducible if it is not replicable. Releasing available HAR algorithms and open-weight HAR models increases the replication and accessibility of researchers' work.

3 Related Literature: Prior Related HAR Surveys and Reviews

To motivate the need for this review, we compare our review to the most closely related existing surveys and reviews. Because our review investigates comparability, availability, and reproducibility in the wrist-accelerometer-only (WAO) human activity recognition (HAR) literature, we searched for reviews of the HAR literature that focused on wrist-worn wearable data, accelerometer only data, open source practices, or replication and reproduction. In addition to searching the reviews we found while screening, we retrieved reviews by searching for papers in Google Scholar[5] with the keywords `("review" OR "survey") AND (wrist OR watch OR wearable OR device) AND activity`. We summarize the reviews and surveys most closely related to our work in Table 1. The related reviews/surveys are limited and do not capture the breadth of reviews/surveys of the general HAR literature because our review focuses on WAO HAR (needed to analyze existing large wrist-accelerometer-only health datasets). We are unaware of any review or survey that covers the same scope as ours.

By constraining our data source, sensor location, and classification tasks to activities relevant to behavioral health researchers (see the "out-of-scope classification task" (**`OOS CT`**) exclusion criterion in Sec. 4.2.2), we explore how researchers in this field report and compare their novel HAR algorithms as well as the transparency within their protocols and supplemental materials. These topics are crucial to the replicability and reach of HAR research, and our review addresses some of the gaps in existing reviews while also corroborating and extending upon their findings.

[3]We define *replicability* in terms of the full article, not a single algorithm within the article. If the authors of a HAR article describe multiple HAR algorithms, but not all of these algorithms can be used to train a HAR model, the article is not replicable, although some of the algorithms may result in usable HAR models.

[4]Like replicability, we define *reproducibility* in terms of an article as a whole. Although an independent researcher may be able to obtain computationally consistent models with *some* of those in the original work, an article is not reproducible unless *all* resulting models are computationally consistent.

[5]www.scholar.google.com/.

Table 1. Known surveys and reviews of HAR literature that have a scope that overlaps with topics related to at least one of the main focuses of this review: wrist-based or accelerometer-only HAR, open-source code and resources, and transparency that leads to more replicable and reproducible (R&R) HAR research. We are unaware of any review with the same scope and results as ours.

Review	Wrist	Accel.	Code	R&R	Description
[106]	✓	✓			Acceleration based HAR using various wearable devices.
[190]		✓			Wear locations and methodologies for HAR in-the-wild.
[177]			✓	✓	Impact of location and modality on smartphone-based HAR. Develops a taxonomy of replication in HAR research.
[116]		✓	✓		Self-supervised learning for accelerometer-based HAR. Discusses the need for more open-source foundational HAR models.
[107]	✓				Methodologies for smartwatch- and smartphone-based HAR, including sensor types and classification tasks.
[173]		✓			Taxonomy of accelerometer-based HAR, discussion of the implications of online versus offline learning in HAR.
[12]	✓				General applications of wrist-worn wearables.
[16]		✓			Accelerometer-based HAR techniques, algorithms, and other factors that impact performance.
[114]	✓	✓			Uses of WAO HAR systems in large-scale studies and randomized trials.
Ours	✓	✓	✓	✓	Comparability, availability, and replicability of WAO HAR algorithms and models.

Column Definitions: **Wrist** - this review included a section on publications using wrist-worn devices for HAR, **Accel.** - this review only considered accelerometer-only-HAR publications, **Code** - this review included a section discussing open-source resources provided by HAR publications, and **R&R** - this review included a section on replications/reproduction of HAR publications.

4 Method: Search and Retrieval, Inclusion Criteria, and Data Extraction

According to the PRISMA guidelines 2020 [150], we describe the search and retrieval methodology, eligibility criteria, and the data extraction protocol used in this review.

4.1 Search and Retrieval

We focused our review on if, and particularly how, the engineering community is aiding the health community with the task of analyzing the wrist-worn accelerometer data collected from large scale surveillance studies. We conducted our search and retrieval using four peer-reviewed HAR publication engineering venues: ACM Digital Library (DL)[6], IEEE Xplore[7], Science Direct[8], and the MDPI Sensors journal[9] as well as PubMed[10]. We included PubMed because some authors publish their WAO HAR work in health venues, offering more visibility to the health surveillance researchers who could use these WAO HAR models.

In each database, we queried the abstracts of publications for a set of search terms. We designed our search terms to capture 1) the problem, i.e., human activity recognition, and 2) the scope of the data, i.e., wrist-worn

[6] www.dl.acm.org/.
[7] www.ieeexplore.ieee.org/Xplore/home.jsp.
[8] www.sciencedirect.com/.
[9] www.mdpi.com/journal/sensors.
[10] www.pubmed.ncbi.nlm.nih.gov/.

acceleration data. When refining our search terms, we chose to ensure a wide breadth of literature was covered. As such, to capture the problem in our search terms, we used (`“activity recognition” OR “activity detection” OR “activity classification”`). Because “human” is not often specified in the literature, we reduced our main keyword to “activity recognition” and added two synonyms, `“activity detection”` and `“activity classification,”` to capture linguistic variations. Additionally, in initial searches, we found that limiting the search terms to (`“wrist” OR “watch”) AND (accelerometer)` resulted in many missed articles because sometimes abstracts do not specify the types of data their HAR models are trained on, or they evaluate multiple data sources from multiple sensors. Thus, we significantly broadened our search terms to include any potentially wearable sensor that could be placed on the wrist. On May 8th, 2026[11], we searched all our databases using the following search terms (examining the paper abstract[12]): `(“activity recognition” OR “activity detection” OR “activity classification”) AND (accel* OR IMU OR sensor OR watch OR phone)`.

Once retrieved from the databases, two authors independently screened each article using Covidence[13]. To be included in the review, all articles had to pass a title and abstract screen, and their full text had to meet all inclusion criteria. The co-authors' title and abstract screening resulted in a Cohen's Kappa agreement of $\kappa = .81$ (“almost perfect” agreement [105]). To ensure every article met the inclusion criteria and was screened properly, any article with unclear or ambiguous writing during the full-text screen was additionally screened by a third co-author. Each ambiguous article was then discussed by one of the original screening authors and the third screener; they together agreed upon the inclusion of articles or the reason for exclusion.

4.2 Eligibility Criteria

Any article eligible for inclusion in this review met all of the inclusion criteria and none of the exclusion criteria. The final eligibility criteria were agreed upon by all authors before screening.

4.2.1 Inclusion Criteria. To be included in our review, the article must:

(1) Be written in English.
(2) Summarize at least one algorithm to recognize an individual doing activities health researchers would like to analyze. In particular, we focus on algorithms capable of recognizing at least two distinct 1) physical activities, including at least one clear ambulation activity (e.g., walking) or a clear sedentary behavior (e.g., lying down or sitting) *or* 2) estimated physical activity intensity levels (i.e., sedentary, light, moderate, vigorous).
(3) Evaluate at least one summarized algorithm[14] on a single wrist-accelerometer-only data stream.

4.2.2 Exclusion Criteria. We ***excluded*** articles if they met any of the following criteria:

(1) **`IRRELEVANT`**: The article is not about human activity recognition (i.e., using a machine learning or deep learning model to classify a set of activities). Articles that do not use a HAR model (e.g., instead use cutpoint/thresholding methods) or do not describe the evaluation of a HAR model were also classified as **`IRRELEVANT`**.
(2) **`NON-ARTICLE`**: The article is published as a poster, workshop paper, extended abstract, or tutorial demo. These types of articles are less likely to have gone through peer review before publication and are more likely to be a “work-in-progress” without rigorous evaluation acceptable to health researchers. Such articles

[11]We completed our initial search on August 7th, 2024. Because HAR is, in general, a fast-moving field, and systematic reviews take time, to keep this review up-to-date at the time of publication, we completed a final search for articles on May 8th, 2026. We performed identical searches each time and merged the data from both searches. The results of this paper include an analysis of all the articles we retrieved during our final search on May 8th, 2026.

[12]PubMed does not allow for abstract only searches, so we searched using the “Title/Abstract” filter.

[13]www.covidence.org/.

[14]This evaluated algorithm must meet inclusion criterion (2).

are thus deemed outside the scope of our review, which is intended to identify HAR models that health surveillance researchers might trust and readily use.

(3) **`SURVEY/REVIEW`**: The article is a survey or review of existing work in HAR and does not introduce or describe results of a HAR algorithm.[15]

(4) **`OOS CT`** (out-of-scope classification task): The article describes an algorithm designed to classify activities that health researchers would not normally use to analyze free-living data. Specifically, the described algorithms are capable *only* of classifying:

  (a) Sub-activities within a *single* sport (e.g., different swimming strokes or basketball moves).

  (b) Sub-activities within a *single* daily activity[16] (e.g., an article describing an algorithm trained to detect "buckling seat belt" and "turning steering wheel" and "driving" is excluded but an algorithm capable of identifying the above activities as well as additional activities like "walking" and "sitting" is included. Additionally, articles describing algorithms capable of classifying only different gaits or walking speeds are excluded).

  (c) Upper body movements (e.g., arm/hand gestures).

  (d) Lower body movements (e.g., leg raises or side steps).

(5) **`NON-WEARABLE`**: The publication described a HAR algorithm that used data from non-wearable sensors, such as cameras.

(6) **`NOT WAO`**: The publication describes and only uses wearable data that is not from a wrist-worn accelerometer.

(7) **`INSUFFICIENT DETAIL`**: The article did not include the necessary details (e.g., activity types or sensor locations) to be deemed included. Each of these articles was screened by at least two authors who both agreed there was insufficient detail in the writing of the article.

(8) **`NOT AVAILABLE`**: The article was either not available, retracted, or not written in English.

4.3 Data Extraction Protocols

All three authors involved in the screening process performed data extraction according to an agreed-upon, written protocol (attached in our supplemental materials). For each included article, two authors independently extracted information about the wrist-accelerometer-based datasets used (e.g., name, sensor location(s), number of participants), data pre-processing techniques (e.g., window size and, if applicable, features extracted), HAR model information (e.g., architecture type and language/software used for implementation), training and evaluation processes (e.g., evaluation metrics, performance, cross-validation/train-test split), and the availability of the algorithm or model used in the article (e.g,. if code, or a functioning link to code, is included in the supplementary material of the article). We randomly assigned each article to two authors for data extraction. After extraction, both authors who extracted data for a given article discussed and resolved any disagreements in the data extraction. Any information that neither author could find during extraction is marked as "N/A" or "-" for the rest of this paper and in our supplemental material. The results of the data extraction for this review are presented in Section 5. An unabridged copy of all data extracted from each article is available as a table in our supplemental material (https://anonymous.4open.science/r/wao-har-systematic-review-1663/, anonymized for submission).

4.3.1 Replication and Reproducibility. During data extraction, we also graded each article as `likely replicable` or `likely` **`not`** `replicable`. Each article we also graded as `likely reproducible` or `likely` **`not`** `reproducible`. The grade for each article was established using the body of the paper and associated supplementary materials (including codebases or repositories) provided by the authors. We considered supplemental material only when it

[15]We examined the survey/review articles we excluded (see Sec. 3) but we did not use them to further search for citations of articles that might qualify for inclusion in our review.

[16]"Daily activities" include activities such as driving, cooking/food preparation, and working out (in a gym).

was discussed/linked in the body of the paper or provided on the publisher's website; we did not use a search engine to find supplemental material if the publisher or authors did not explicitly state it was provided.

We graded an article as `likely` **`not`** `replicable` because it met at least one of the following criteria:

(1) **`IP`** (insufficient parameters): the article lacks specific detail on algorithm parameters (e.g., kernel, number of neighbors, or learning rate) that would require future researchers to make a judgment call. These articles may cite prior work that may or may not include the specific parameters.[17]
(2) **`IF`** (insufficient feature set): the article lacks detail on the exact feature extraction methodologies, or feature set, that would require future researchers to make a judgment call.
(3) **`IS`** (insufficient segmentation protocol): the article lacks detail on the exact segmentation methodologies that would require future researchers to make a judgment call. This could apply to either window length or window overlap.

We graded an article as `likely` **`not`** `reproducible` because it met at least one of the following criteria:

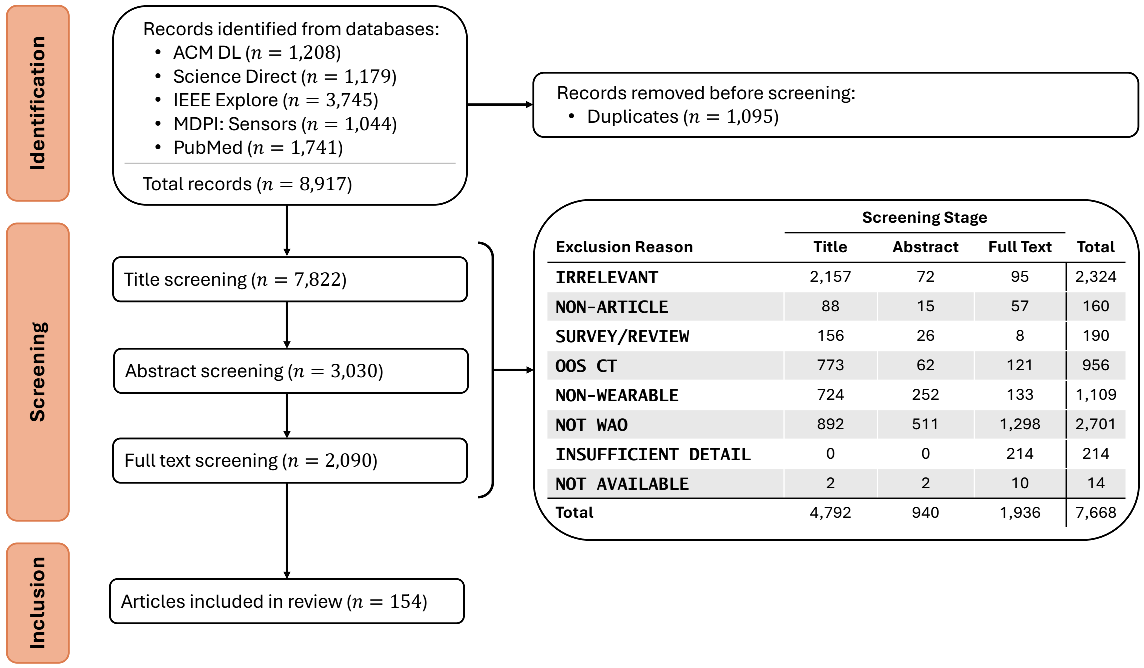
(1) **`DNA`** (data not available): the article used data for training/evaluation that were not explicitly noted as publicly available.
(2) **`TTS`** (train/test split): the article lacks detail on the exact train/test splits used to train and evaluate models that would require future researchers to make a judgment call.
(3) **`ANR`** (algorithm not replicable): the article used an algorithm that we do not believe could be implemented from the algorithm description, or describes results that rely on an explicit seed that was not reported.

Although two authors, both with experience in re-implementing existing algorithms and human activity recognition, agreed on the replication/reproduction grade for each article, we recognize this grade is subjective. Articles we deemed as likely *not* replicable/reproducible perhaps are. For example, if the authors of an article used default parameters but did not specify as such, their work may be replicable and/or reproducible. Although our replication/reproduction grades have such limitations, we use these grades as a starting point to foster a conversation about the state of transparency of reporting, which may impede replication and reproduction of WAO HAR research.

5 Results

Our initial search of five databases (described in Sec. 4.1) returned a total of 8,917 articles. We removed 1,095 duplicated articles before screening the remaining articles and we ultimately included in 154 articles in our review. We summarize the results of our screening protocol in Fig. 2. From each included article we extracted 33 pieces of information, not including metadata, from the 154 articles included in this review. We additionally graded each article as likely replicable and/or reproducible (we detail our extraction protocols in Sec. 4.3) and categorized the article based on the main contribution. We summarize the key results of this extraction in by contribution area in 2. In the following sections we describe how these data address our research questions. We have included all data extracted from these articles in our supplemental material.

[17]We recognize that some journals impose word count or page limits, which may prevent authors from describing the details of their work in the body of the article. However, some citations lead to articles containing multiple possible parameters and architectures, making it difficult, if not impossible, to distinguish what the authors did. When sufficient supplemental material was provided, articles were not graded as **`IP`**, even if the body of the article did not include these details.



Exclusion Reason	Title	Abstract	Full Text	Total
IRRELEVANT	2,157	72	95	2,324
NON-ARTICLE	88	15	57	160
SURVEY/REVIEW	156	26	8	190
OOS CT	773	62	121	956
NON-WEARABLE	724	252	133	1,109
NOT WAO	892	511	1,298	2,701
INSUFFICIENT DETAIL	0	0	214	214
NOT AVAILABLE	2	2	10	14
Total	4,792	940	1,936	7,668

Fig. 2. The results of our screening protocol (described in Section 4.1), following the PRISMA 2020 guidelines [150]. We screened 7,822 articles and ultimately included 154 in our review.

Table 2. The reviewed articles, sorted by main contribution area, highlighting differences in the WAO HAR algorithms, evaluation protocols, and open source practices.

	R&R		Open...			Validation				Evaluation			Wrist Side					No. Activities		
	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
Algorithmic Changes (e.g., architecture or preprocessing)																				
[123]								✓	✓		✓									✓
[124]		✓						✓	✓		✓	✓								✓
[21]					✓		(✓)		✓		✓									✓
[25]					✓		✓		✓		✓					✓		✓	✓	
[28]	✓	✓	(✓)	✓	✓		✓		✓			✓							✓	
[37]					✓			✓	✓	✓							✓		✓	
[160]									✓		✓			✓						✓

continued on the next page

	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
[42]								✓				✓		✓					✓	
[43]	✓	✓		✓	✓		✓	✓	✓			✓		✓		✓			✓	✓
[50]	✓	✓		✓	✓			(✓)	✓		✓									✓
[48]		✓							✓			✓				✓		✓		
[53]					✓			✓	✓	✓									✓	
[62]					✓			✓	✓		✓		✓					✓		
[66]					✓		✓	✓	✓			✓	✓					✓		
[67]		✓			✓		✓	✓		✓									✓	
[68]					✓			✓	✓	✓										✓
[54]									✓		✓									✓
[69]										✓						✓		✓		
[74]	✓	✓		✓	✓				✓			✓								✓
[77]					(✓)		✓	✓	✓			✓	✓				✓	✓		✓
[76]		✓			✓		✓				✓	✓				✓		✓		
[79]	✓	✓			✓		✓		✓		✓								✓	
[84]		✓			✓				✓		✓							✓		
[86]								✓												✓
[91]	✓	✓			✓				✓			✓					✓		✓	
[92]				(✓)	✓		✓	✓		✓								✓		
[93]		✓			✓			✓	✓	✓								✓		
[94]					✓				✓			✓								✓
[99]		✓		(✓)				✓										✓		
[108]		✓						✓	✓			✓	✓					✓		
[110]					✓		✓					✓						✓		
[112]								✓	✓	✓									✓	
[113]							✓			✓				✓						✓
[117]		✓			✓			✓		✓									✓	
[115]					✓					✓						✓		✓		
[119]					✓		✓	✓	✓				✓					✓		
[118]	✓	✓	✓	✓	✓		✓	✓	✓	✓									✓	✓
[120]		✓			✓		✓	✓	(✓)											✓
[121]					✓		✓		✓	✓			✓					✓		
[122]		✓		✓	✓				✓	✓						✓		✓		
[87]		✓			✓		✓	✓	✓		✓					✓		✓		
[128]								✓	✓		✓		✓					✓		
[129]								✓	✓	✓									✓	
[132]		✓						✓				✓				✓				✓
[135]					✓				✓		✓					✓		✓		
[136]	✓	✓			✓		✓		✓			✓				✓		✓		
[133]					✓				✓		✓					✓			✓	

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	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
[138]					✓			✓	✓							✓		✓		
[137]					✓			✓	✓		✓		✓					✓		
[134]					✓				✓									✓		
[139]					✓			✓	✓		✓		✓						✓	
[142]					✓				✓		✓					✓		✓		
[143]								✓	✓		✓				✓			✓		
[145]					✓		✓	(✓)	(✓)		✓		✓						✓	
[146]					✓			✓	✓			✓						✓		
[155]					✓			✓	✓		✓							✓		
[166]					✓			✓	✓										✓	
[167]		✓						✓			✓								✓	
[169]		✓			✓		✓	✓	✓										✓	
[170]					✓			(✓)	✓			✓							✓	
[171]		✓					✓		✓						✓					✓
[175]					✓				✓		✓	✓					✓		✓	
[179]								✓				✓		✓					✓	
[180]							✓	✓	✓	✓			✓						✓	
[181]				(✓)	✓			✓			✓				✓				✓	
[182]				(✓)	✓			✓		✓	✓	✓							✓	
[188]					✓			✓		✓			✓						✓	
[185]								✓												✓
[186]		✓						✓	✓			✓			✓				✓	
[187]		✓						✓	✓			✓			✓				✓	
[199]		✓						✓	✓	✓									✓	
[201]					✓			✓		✓						✓		✓		
[204]	✓	✓		✓	✓		✓	✓	✓	✓										✓
[205]		✓		✓	✓		✓	✓	✓	✓									✓	
[206]		✓		✓	✓			(✓)		✓									✓	
[207]								✓	✓										✓	
Algorithmic Changes (Summary)																				
Count (76)	9	30	1	9	50	0	23	47	54	23	25	22	12	5	5	15	4	29	32	19
% (of 76)	(12)	(39)	(1)	(12)	(66)	(0)	(30)	(62)	(71)	(30	(33)	(29)	(16)	(7)	(7)	(20)	(5)	(38)	(42)	(25)
Personalized HAR																				
[34]		✓		✓	✓				✓		✓							✓		
[72]					✓		✓	(✓)	✓		✓		✓					✓		
[125]	✓	✓			✓		(✓)					✓				✓				✓
[164]								✓	✓			✓								✓
[176]	✓	✓			✓	✓	✓	✓				✓	✓							✓

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	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
[198]					✓		✓			✓			✓						✓	
Personalized HAR (Summary)																				
Count (6)	2	3	0	1	5	1	3	2	3	1	2	3	3	0	0	1	0	2	1	3
% (of 6)	(33)	(50)	(0)	(17)	(83)	(17)	(50)	(33)	(50)	(17)	(33)	(50)	(50)	(0)	(0)	(17)	(0)	(33)	(17)	(50)
Data Collection Methodology (e.g., sensor location or modality)																				
[14]		✓			✓				✓								✓	✓		
[17]								✓				✓				✓				✓
[7]		✓			✓			✓			✓	✓		✓						✓
[103]		✓						✓	✓			✓	✓							✓
[19]		✓										✓						✓		
[29]	✓	✓		✓	✓			(✓)	✓			✓	✓					✓		
[44]								✓	✓		✓				✓				✓	
[49]	✓	✓		✓	✓		✓	✓	✓		✓						✓	✓		
[51]								✓			✓		✓					✓		
[63]		✓						✓				✓				✓				✓
[90]											✓					✓				✓
[98]		✓						✓	✓			✓				✓			✓	
[126]	✓	✓		✓	✓			✓				✓				✓				✓
[140]					✓						✓									✓
[141]					✓			✓	✓	✓										✓
[151]		✓						✓	✓		✓								✓	
[154]	✓	✓	✓	✓	✓	(✓)	(✓)	✓				✓			✓			✓	✓	✓
[161]					✓			✓	✓			✓					✓		✓	
[165]					✓		✓	✓	✓			✓				✓				✓
[168]	✓	✓			(✓)			✓	✓		✓					✓		✓		
[183]	✓	✓		✓	(✓)			(✓)				✓					✓	✓		
[189]		✓		(✓)				✓			✓			✓					✓	
[197]									✓		✓	✓				✓		✓		
[203]		✓						✓	✓		✓						✓			✓
Data Collecton Methodology (Summary)																				
Count (24)	6	15	1	5	10	0	2	17	13	1	11	12	3	2	2	8	5	9	6	11
% (of 24)	(25)	(63)	(4)	(21)	(42)	(0)	(8)	(71)	(54)	(4)	(46)	(54)	(13)	(8)	(8)	(33)	(21)	(38)	(25)	(46)
Generalized HAR																				
[26]					✓				✓		✓							✓		
[36]					(✓)		✓		✓		✓		✓			✓	✓		✓	✓
[39]					✓		✓		✓	✓						✓	✓	✓		✓
[40]		✓			✓		✓										✓		✓	✓
[64]		✓					(✓)		✓	✓				✓				✓		

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	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
[78]		✓		✓		(✓)	(✓)	✓								✓		✓		
[80]					✓	✓	✓			✓			✓					✓		
[85]					✓	✓	✓				✓								✓	
[89]		✓						✓	✓			✓			✓					✓
[71]					✓		✓		✓			✓	✓			✓		✓	✓	✓
[184]	✓	✓			✓	✓	✓				✓								✓	
[194]		✓							✓		✓		✓						✓	
[200]	✓	✓	(✓)	✓	✓	✓	✓		✓		✓									
Data Collection Methodology (Summary)																				
Count (13)	2	7	0	2	8	4	8	2	8	3	6	2	4	1	1	4	3	6	6	5
% (of 13)	(15)	(54)	(0)	(15)	(62)	(31)	(62)	(15)	(62)	(23)	(46)	(15)	(31)	(8)	(8)	(31)	(23)	(46)	(46)	(38)
Online or Real-time HAR																				
[15]								✓				✓			✓			✓		
[20]								✓	(✓)	✓									✓	
[27]							✓	✓			✓	✓			✓					✓
[45]					✓					✓			✓					✓		
[52]								✓	✓		✓		✓					✓		
[193]					✓					✓								✓		✓
[58]					✓			✓		✓										✓
[57]					✓		✓				✓									✓
[60]									✓	✓						✓			✓	
[65]				(✓)	✓															✓
[75]																		✓		
[96]					✓		✓	✓	✓		✓								✓	
[95]								✓		✓						✓			✓	
[97]					✓		✓	(✓)			✓							✓		
[149]	✓	✓			✓					✓							✓	✓		
[159]					✓			✓			✓									✓
[162]									✓		✓	✓								✓
[172]					✓			✓	✓	✓									✓	
[202]								✓		✓									✓	
Online or Real-time HAR (Summary)																				
Count (19)	1	1	0	0	10	0	4	10	5	9	7	3	2	0	2	2	1	7	6	7
% (of 19)	(5)	(5)	(0)	(0)	(53)	(0)	(21)	(53)	(26)	(47)	(37)	(16)	(11)	(0)	(11)	(11)	(5)	(37)	(32)	(37)
HAR for a niche population or activity set																				
[8]		✓						✓	✓			✓								✓
[9]								✓	✓					✓						✓
[10]	✓	✓	✓	✓	✓			✓				✓								✓

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	Reproducible	Replicable	Weights	Algs.	Data	Cross DS	Mult. DS	Class Wise	Baseline	Single Split	*k*-fold CV	LOSO CV	R	ND	L	D	B	10+	6-9	2-5
[11]	✓	✓	✓	✓	✓			✓				✓		✓						✓
[30]					✓			✓	✓	✓						✓		✓		
[38]		✓						✓	✓	✓				✓						✓
[70]									✓		✓			✓					✓	✓
[73]								✓	✓	✓							✓			✓
[104]	✓	✓		✓	✓			✓	✓		✓					✓		✓		
[111]								✓	✓			✓		✓						✓
[127]	✓	✓		✓	✓			✓				✓				✓				✓
[144]									✓	✓										✓
[156]		✓						✓			✓									✓
[163]		✓						✓	✓			✓				✓				✓
[195]	✓	✓			✓				✓		✓						✓		✓	
Niche HAR (Summary)																				
Count (15)	5	9	2	4	6	0	0	12	11	4	4	6	0	5	0	4	2	2	2	12
% (of 15)	(33)	(60)	(13)	(27)	(40)	(0)	(0)	(80)	(73)	(27)	(27)	(40)	(0)	(33)	(0)	(27)	(13)	(13)	(13)	(80)
Misc.																				
[101]					✓		✓		✓	✓	✓	✓	✓					✓		
Misc. (Summary)																				
Count (1)	0	0	0	0	1	0	1	0	1	1	1	1	1	0	0	0	0	1	0	0
% (of 1)	(0)	(0)	(0)	(0)	(100)	(0)	(100)	(0)	(100)	(100)	(100)	(100)	(100)	(0)	(0)	(0)	(0)	(100)	(0)	(0)
All Included Articles (Summary)																				
Count (154)	25	65	4	21	90	5	41	90	95	42	56	50	25	13	10	34	15	56	53	57
% (of 154)	(16)	(42)	(3)	(14)	(58)	(3)	(26)	(59)	(62)	(27)	(37)	(32)	(17)	(8)	(7)	(22)	(10)	(36)	(34)	(37)

Symbols: ✓ - this article meets the criteria for the column heading, (✓) - this article *partially* meets the criteria for the column heading (i.e., a link may be broken or includes class-wise results for a non-WAO HAR model). *Abbreviations*: **B** - both wrists, **D** - dominant, **L** - left, **ND** - non-dominant, **R** - right, **CV** - cross validation, **LOSO** - leave-one-subject-out, **Algs.** - algorithms, **Mult.** - multiple, and **DS** - dataset.

5.1 Developing and Comparing WAO HAR Algorithms (**RQ1**)

To address **RQ1** (i.e., how researchers develop and compare novel and existing WAO HAR algorithms), we investigated the contribution areas, commonly used datasets, and evaluation/validation protocols reported in each of the included articles.

RQ1 Results: Across and within the six main contribution areas, we found no clear consensus on the datasets used, classification tasks, sensor wear location (i.e., which wrist), or evaluation/validation protocols. Most articles (113 articles (73%)) train and evaluate their proposed algorithm on a single dataset and, in the case of generalized models, only 2 articles (15%) use LOSO CV, meaning many publications may be inadvertently overstating their results as they apply to a more niche subset of the population and activities than their contribution is framed as. Further, across our included articles, only 90 articles (59%) report classwise results and only 95 articles (62%) report how their algorithms compare to existing algorithms, potentially making it difficult to decipher if this is the best algorithm to apply to certain tasks. The general *lack of standardization makes WAO HAR algorithms and models difficult to directly compare or for health researchers to establish which may work best for their task.*

5.1.1 Contribution Areas in WAO HAR Research. We identified six main contribution areas in the WAO HAR literature: algorithmic changes, personalization, improved data collection methodologies, generalization, moving towards online/real-time HAR, and HAR for niche populations or activities. A total of 76 articles (49%) define the main contribution of their article to be an algorithmic change (i.e., the authors propose a "better" architecture or new preprocessing strategy). Another 24 articles (16%) focused on problems related to data collection, such as specific sensor wear locations or the benefit of collecting additional modalities. Only 13 articles (8%) make contributions specifically towards WAO HAR algorithms that will be robust on a general population, an open question necessary to solve for health researchers analyzing population wide physical activity trends.

5.1.2 Commonly Used WAO HAR Datasets. In our review, we found researchers used 27 different publicly available datasets containing wrist accelerometer data. We report the datasets used by more than one included article in Table 3. Additionally, we found the authors of 82 articles (53%) collected their own datasets for their research but only 14 of those datasets were released to the public for others to use.[18]

Table 3. Datasets that contain wrist accelerometer data used by more than two articles included in this review. Many articles used data from multiple datasets. Additional wrist accelerometer datasets used by articles in this review and their availability can be found in our supplemental materials.

Dataset	# Articles	*N*=	# Acts	Wrist Side	Link
WISDM 2019 [196]	15	51	18	Right	UCI MLR
PAMAP2 [157]	13	9	18	Right	UCI MLR
HHAR [175]	7	9	6	Both	UCI MLR
MHEALTH [22, 23]	6	10	12	Right	UCI MLR
RealWorld [181]	6	15	8	Left	University of Mannheim
WHARF [32]	6	16	14	Right	UCI MLR
Opportunity [158]	4	4	6	Dominant	UCI MLR
Mannini [126]	4	33	26	Right	Northeastern University
DaLiAc [109]	3	23	13	Right	FAU
DHA [102]	3	2	11	Dominant	Dropbox
DSADS [18]	3	8	19	Both	UCI MLR
Self-collected	82	Variable	Variable	Variable	N/A

[18]Notably, many of the links to these datasets are broken.

Even though many researchers used the same public data, the manner they used these data differed. For example, Roy et al. [159], Farag et al. [67] and Hnoohom et al. [84] all trained and evaluated WAO HAR algorithms using the WISDM (2019) dataset [196]; however, the algorithm by Hnoomhom et al. [84] classifies all 18 activities labeled in the dataset, the algorithm by Farag et al. [67] classifies six activities from the original 18 activities, and the algorithm by Roy et al. [159] classifies three energy levels made by merging activities in the WISDM dataset by assumed activity level.[19] Although specific use may differ, public data helps to ensure the work is reproducible; however, researchers wanting to study specific problems are limited by the scope of activities in the already publicly available data.

In addition to limiting what activities models can learn, publicly available datasets may limit the sensor locations models are capable of recognizing activity from. We scoped our review to wrist-accelerometer-only HAR, however, gesturing as well as dominance can impact the signals collected by wrist-worn devices [130]. That said, the majority of the publicly available datasets used right/left denominations instead of focusing on the participants' hand dominance. Further, although a sensor placed on the dominant wrist may better capture gesturing, most individuals wear a watch (which may contain an accelerometer) on their non-dominant wrist. Thus, to least disturb participants in their daily lives while still being able to objectively measure their physical activities, health researchers may be inclined to prefer models that focus on non-dominant wrist accelerometer data. To that end, in this review we found 34 articles (22%) detailed a WAO HAR algorithm using data from the dominant wrist, and another 25 articles (17%) described using data from the right wrist. Only 38 articles (25%) detailed WAO HAR algorithms that used data from the left, non-dominant, and both wrists (separately). The difference in the description of sensor wear location in public datasets in combination with how sensors may be worn in real-world conditions limit the research questions that can be asked and answered and the generalization of the work. More realistic and complex physical activity datasets are needed to train, evaluate, and validate WAO HAR models that can be used as analysis tools by health researchers.

5.1.3 Evaluation and Validation Strategies. Most health researchers looking to use a WAO HAR model on their data care about (1) how a model might perform on their dataset (i.e., if a model is trained on a specific dataset, does that performance generalize?), (2) how a model performs on specific activities within their dataset (i.e., if a model performs well overall, does it perform well on the activities the health research cares about?), and (+3) if the model is built for the population that they care about for (i.e., is this model for "healthy adults" or a different, specific demographic?). Although we have considered personalized models within the scope of this review, these health researchers analyzing large epidemiological data, at present, care more about analyzing movements at a population-wide scale than about a unique individual's activities or movements. We considered these factors when analyzing the evaluation strategies and metrics WAO HAR researchers use."

Although validation is important to understand how a trained model may perform on novel data, we identified only five articles (3%) that demonstrate the ability of their WAO HAR model to generalize to new datasets via a cross-dataset validation. We found 41 articles (26%), however, that had trained and evaluated their algorithm across multiple datasets; the remaining 114 articles (74%) chose to report the results of their WAO HAR model on a single dataset and did not validate their model across datasets. Not validating WAO HAR models or reporting the results of algorithms on multiple datasets highlights a misalignment between the goals of WAO HAR researchers and those of potential end users in developing and using these models.[20] Further, when considering generalized

[19]Roy et al. [159] defined these classes as follows: "The sedentary class encompasses stationary activities such as standing and sitting. Moderately active activities involve ambulatory motion, primarily represented by walking. Finally, the active class pertains to more vigorous physical activities, such as jogging."

[20]We consider the authors of a paper to evaluate a HAR algorithm on datasets if the same algorithm is trained and tested on multiple datasets, regardless of data source (e.g., gyroscope data or a thigh-worn accelerometer). These additional datasets may not meet the criteria defined as an "in-scope classification task" (see **OOS CT** in Sec. 4.2.2).

models, leave-one-subject-out cross-validation (LOSO CV) should be used to best assess the true reliability of the model [31, 101], however, we found that only two articles that considered generalized WAO HAR (15%) used leave-one-subject-out cross-validation (LOSO CV). Because the authors did not use LOSO CV, they may be overstating their performance and inadvertently misleading potential end users of the models. Additionally, many end users of WAO HAR models may want to understand how models perform on specific activities, which requires authors to report class-wise results either in a table or by using a confusion matrix. We found 90 articles (59%) report the class-wise performance of their models, allowing end users to compare the performances of different models on the activities that they care about.

Lastly, to highlight the benefit on their model, WAO HAR researchers may want compare the performance of their model against others. Assuming they use the same evaluation strategy (e.g,. split and metric), training and evaluating multiple algorithms on the same datasets allows for the most direct comparison. We found 95 articles (62%) directly compared their solution with an existing algorithm. Including these comparisons make it easier for anyone to understand the pros and cons of choosing different algorithms with respect to overall and classwise performance.

Developing and deploying a WAO HAR model that can be used to analyze existing population-wide health datasets requires a deep understanding of what activities a model can reliably detect from a specified sensor-wear location, and how that model may perform on novel datasets. Because the classification task can differ between models and algorithms can be built to address different challenges (e.g., personalized models versus generalized models), some models perhaps should not be directly compared. That said, because there is a lack of standardization in evaluation and validation it is difficult to compare how various WAO HAR models perform on existing, labeled datasets, let alone their limitations when being applied to unlabeled, population-scale datasets.

5.2 Publicly Available WAO HAR Algorithms and Open-weight Models (**RQ2**)

To address **RQ2** (i.e., what WAO HAR models are currently available to health researchers), we identify and summarize the papers whose authors released open-weight HAR models.

RQ2 Results: We identified *only four articles (3%) whose authors released a ready-to-use, trained WAO HAR model* to accompany the article (we assess the usability of these models in Sec. 6.1.2). All authors released at least one open-weight model trained only on data collected in a laboratory or semi-naturalistic setting, and only two articles (50% of articles releasing open-weight models) released an open-weight WAO HAR model trained on data from adults. Because many of these models were not trained on free-living data from adults, they may not be suitable to use for analysis on the data that has been collected in large-scale health surveillance studies of the adult population [64, 147]. Only Potter et al. [154] released an open-weight model trained on free-living data from adults that may be suitable for this purpose, but further evaluation is needed because they did not evaluate their models' performance across datasets. Additionally, we identified only *21 articles (14%) in total that made their HAR algorithm available by publishing their source code.*

Lu et al. [118] released two open-weight WAO HAR models to accompany their publication on a pipeline investigating the impacts of transforming accelerometer data into a recurrence plot before doing activity recognition. These two models are from training the same algorithm (a ResNet) on two different datasets: ASTRI[21] and WHARF [32], each with their own classification tasks. The model trained on the ASTRI data is capable of classifying five activities (walking, standing, squatting, sitting, and lying) using data from either wrist. And, the model

[21]The ASTRI dataset can be found at https://github.com/lulujianjie/robust-single-accelerometer-based-activity-recognition-using-modified-recurrence-plot/tree/master/Datasets/MotionData.

trained on the WHARF dataset is capable of classifying seven activities (climb stairs, drink glass, getup bed, pour water, sit down, stand up, and walk) from the right wrist. To demonstrate the robustness of their algorithm, Lu et al. compared their novel models against seven other baseline HAR algorithms; their released models slightly outperformed an SVM, the next best performing algorithm, by 2.4% on the ASTRI dataset and by 1% on the WHARF dataset.

Ahmadi et al. (July 2020) [10] released multiple open-weight models based on random forests (RFs) trained on data from various sensor locations (wrist, hip, and ankle) to accompany their work investigating the effects cerebral palsy may have on the task of activity recognition in children. Their released WAO HAR models are capable of classifying three different activity classes: sedentary, standing utilitary movements (e.g., wiping a table), and walking; each model is either personalized within a group[22] or general to a group. Ahmadi et al. (Jul. 2020) did not compare their algorithm against other HAR algorithms or evaluate their algorithm on different datasets (likely because the classification task is uniquely suited to a specific demographic that is only present in the dataset they collected).

Ahmadi et al. (Aug. 2020) [11] released multiple open-weight models based on RFs trained on data from the hip, ankle, and wrist from their work investigating activity recognition in preschool-age children. Their WAO HAR models classify five activity classes: sedentary, light activity and games, moderate to vigorous activities and games, walking, and running. Ahmadi et al. (Aug. 2020) did not compare their algorithm against other HAR algorithms or evaluate their algorithm on different datasets (likely because the classification task is uniquely suited to a specific demographic that is only present in the dataset they collected for this research).

Potter et al. [154] released multiple open-weight models based on RFs trained on accelerometer data from the left wrist and right thigh to benchmark their free-living and laboratory collected physical activity dataset. In total, they released the weights of 22 RF models (half of which are specifically for WAO data) trained on either lab or free-living data. The released models are capable of classifying between five activity classes (Sitting, Standing, Walking, Biking, and Lying Down) and forty-two niche activities. Potter et al. did not compare these algorithms against other HAR algorithms or evaluate their algorithm on different datasets (likely because the purpose of these models was to benchmark a novel dataset).

While it is commendable that the authors of these papers provided open-source code and open-weight WAO HAR models, the dearth of articles that provide usable resources as artifacts of their work is disheartening. Further still, we scoped our review to include articles that describe a WAO HAR algorithm capable of classifying at least one ambulatory or sedentary behavior (see **`OOS CT`** in Sec. 4.2.2)), and while all these models meet that criterion, they also classify many activities health researchers will not find of interest. For example, the ability to recognize "packing/unpacking a box while standing" may come in handy now and again, but such activities are not activities the majority of people spend significant amounts of their time doing and may not need to be considered when examining a large-scale dataset with tens or hundreds of thousands of people from an epidemiological perspective to assess overall behavioral health.

Providing the community with open-weight models and open-source WAO HAR solutions is important to ensuring that others can effectively extend and replicate prior work. Making the code associated with WAO HAR work available, however, is insufficient for the work to be efficiently replicated. Any code or supplementary material should ideally be evaluated and tested by an independent researcher to ensure the resources provided alongside a published article are actually usable (see Sec. 6.1.2). The lack of open-weight models and publicly available codebases in the WAO HAR literature shows a misalignment in the engineering and behavioral health

[22]The authors grouped participants by diagnosis of cerebral palsy at Gross Motor Function Classification System (GNFCS) level I, II, or II.

researchers' goals: *more publicly available, independently validated WAO HAR models are needed to effectively analyze existing large-scale WAO datasets.*

5.3 Replication and Reproduction in WAO HAR Research (**RQ3**)

To address **RQ3** (i.e., what articles are replicable and reproducible), we graded each article as `likely (NOT) replicable/reproducible` and analyzed the results (see Section 4.3.1 for details on the grading protocol). Because this was a grade we assigned to articles and not a piece of information we extracted, it is possible that some articles we did not deem replicable/reproducible are, in fact, just that; however, two authors, both with experience in HAR, independently graded each article, and merged all disagreements.

RQ3 Results: We found the *majority of WAO HAR publications are likely **not** replicable or reproducible.* From the 154 reviewed articles, we graded only 65 articles (42%) as `likely replicable` and only 25 articles (16%) as `likely reproducible`.

5.3.1 Replicable WAO HAR Research. By definition 2.4, a WAO HAR article was graded as `likely replicable` if and only if the authors make the entirety of their algorithm code publicly available or if the authors of the publication describe their process in such detail that someone can identically recreate their protocol without making any assumptions. Using this definition, we graded 65 articles (42%) as being `likely replicable`. We graded only 21 of these articles (32%) as `likely replicable` because the authors made their algorithm available. The authors of an additional six publications either made a portion of their algorithm available (e.g,. just feature extraction) or the link to their code is broken.[23] The most common reason we marked articles as `likely NOT replicable`, affecting 54 articles (35%), was for **`IP`**; the authors did not disclose important parameters of their algorithm's architecture or training protocol.

5.3.2 Reproducible WAO HAR Research. Although we graded slightly less than half of the articles as `likely replicable`, we graded only 25 articles (16%) as being `likely reproducible`. Articles were graded as `likely NOT reproducible` if the original training and evaluation data were not available, the train-test split was not detailed, and/or the publication was not replicable (i.e., someone cannot, without making judgment calls, implement their original algorithm). Of the articles we graded as `likely NOT reproducible`, the most common reason was **`DNA`**; all the original data were not explicitly listed as being publicly available in 60 articles (39%). We graded 56 articles (36%) as **`TTS`**, i.e., `likely NOT reproducible` because of the lack of transparency in reporting the train-test split. Although the authors of 17 articles did not disclose their train-test split at all (Fig. 3), most authors reported their train-test split using a generic term such as an 80-20 split. These generic terms provide a summary of the protocol the authors followed; however, they do not provide enough information for someone to understand exactly what data should be used in training to produce a model computationally consistent to the one in the original publication.

Creating replicable and reproducible articles is difficult, and these challenges are faced across scientific communities (e.g., in psychology [148]). The level of detail required to replicate or reproduce the work of a WAO HAR publication is cumbersome to write and oftentimes not valued by the academic community, who may value different things (e.g., number of publications or citation count). Replication and reproduction, however, allow the community to easily use and compare algorithms so researchers may hopefully converge on the best solution. Although it requires time and effort to provide supplemental materials, including source code, more transparent

[23]For each broken link, we searched for the codebases using Google (www.google.com/). Ultimately, we were able to find one codebase that we have deemed "available," but we are unable to find the other codebases and have noted these instances in our supplemental material.

practices in the WAO HAR community would provide future researchers, in any domain, a better appreciation of and more access to the work.

6 Discussion

Our review underscores the need for more easily comparable, available, and reproducible WAO HAR research. We discuss the road towards more accessible WAO HAR work, the limitations and biases in this review, and potential future work.

6.1 Towards Comparable, Available, and Reproducible WAO HAR

During this review, we found that it is difficult to establish if a WAO HAR algorithm will work for analyzing large-scale health surveillance datasets and are rarely made available. Further, despite being peer-reviewed, the published articles themselves lack transparency, making replication and reproduction difficult. To propose concrete next steps for the WAO HAR research community, we discuss (1) the factors preventing WAO HAR publications from being replicable and reproducible and (2) if the current practices of making model weights open allow non-technical researchers, such as those in behavioral health, to easily use the WAO HAR models. Based on the results of the review, we additionally discuss specific measures the HAR community can take to promote transparency and items HAR researchers should communicate in their work (see Appendix A for an explicit checklist for researchers).

6.1.1 What is Preventing Replication and Reproduction? In our review, we graded only 42% as `likely replicable` and 16% as `likely reproducible`. Although we categorized details preventing replication and reproduction into specific codes, we graded articles as `likely NOT replicable/reproducible` because many different elements lacked clarity or were missing from the paper (Fig. 3).

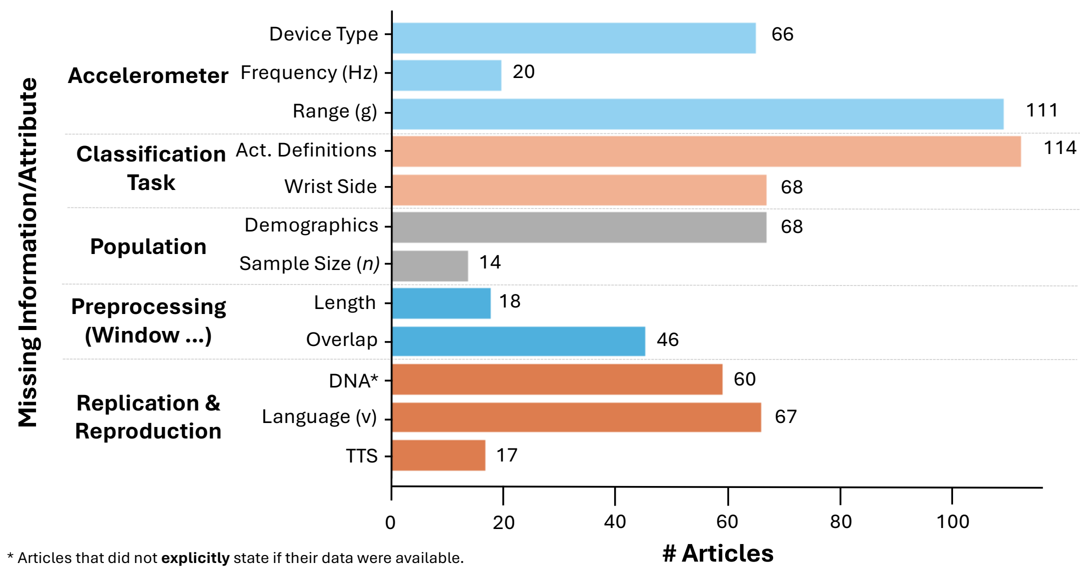

Fig. 3. The included articles are not all replicable or reproducible because of missing information or vague language. Of the 154 reviewed articles, we graded only 65 articles (42%) as `likely replicable` and only 25 articles as (16%) `likely replicable`.

One of the easiest-to-resolve issues that prevents reproduction is to use publicly available data or to release all data associated with a HAR article. Because there are many available wearable-sensor-based HAR datasets [55], collecting a novel dataset is only productive when it allows for the investigation of a niche activity set (e.g., mountaineering activities as in [195]) or a specific population (e.g., pre-school age children as in [11]). Publicly releasing data allows future researchers to use these data for new analyses and provides more transparency to the original work.

In addition to using private data, which directly prevents the reproduction of work, many authors did not include sufficient details that would promote the replication of their work. These details range from information on data segmentation to the data collection methodology, such as the precise wrist a sensor was worn on. Some authors likely omitted this information because their WAO HAR model used publicly available data that should detail this information.[24] Beyond algorithmic opacity, some authors also did not include information crucial for evaluating their models, such as the train-test split. Lack of transparency around evaluation methodology may cause a reader to question whether training data were included in the testing data, or may cause a reader to distrust the results entirely. Precision and transparency when writing about WAO HAR models is critical because when the details of a WAO HAR model cannot be deciphered, the HAR model cannot be applied with certainty to independent, unanalyzed data, preventing the use of valuable models.

Beyond issues that prevent direct replication and reproduction of WAO HAR work, many of the reviewed articles were missing information needed for future researchers to understand their work or analyses. The minority of included articles explicitly defined the specific activities their models could classify, instead using generic terms such as "walking" or "sitting." Although activity definitions may seem pedantic, "walking" may mean "walking on a treadmill at 3 mph," "walking while talking with a friend outside," or "walking while carrying a beverage in hand," all of which produce different accelerometer signals due to movements intrinsic to the activity, such as gesturing [130]. Thus, training a model to identify "walking" in one dataset does not necessarily mean "walking" in another dataset can be easily recognized, especially if the definitions of "walking" are different. Further, many articles neglected to include participant demographics even though demographics may impact how activities are performed and thus a HAR model's performance (e.g., HAR models trained on adults' movements may not generalize to children's movements [111]). Assuming these data are public, this lack of transparency does not impact replication or reproduction; however, this information may alter the applications of the resulting models, and thus should be clearly communicated.

6.1.2 Providing Usable Open-source HAR Algorithms and Models. Increasing the transparency and detail with which HAR researchers report their results does not directly yield more available, ready-to-use WAO HAR models that the community may need. During this review, we identified only four publications that made their WAO HAR models open weight.[25] Make an algorithm open source or a model open weight, however, does not mean that another researcher can use these resources. To verify how easy to use each of these models was, we ran the provided code and summarized any issues in Table 4.

We found that each of the identified WAO HAR models was usable; however, some may be difficult for those without machine learning or programming experience to use. Thus, some may struggle to use these models for analysis of unlabeled, independently collected dataset. We hope the WAO HAR community can learn from these

[24]Although this information may be relayed in the original dataset paper, including it (with proper citations) in WAO HAR work only increases the transparency of the work.

[25]Because of the limitations in our review (Sec. 6.2), it is possible there are more readily available WAO HAR models (e.g., GGIR [191] is an open-source R package for WAO HAR); however, we are unsure where these models may have been published and to which audiences. In the case of GGIR, the authors published their article in the Journal of Measurement of Physical Behavior (https://journals.humankinetics.com/view/journals/jmpb/jmpb-overview.xml), and used terms such as "generating physical activity and sleep outcomes" instead of more typical engineering terms like "activity recognition" that we considered as keywords in our search).

[26]We have emailed the authors of this work and, as of submission, we have not heard a response.

Table 4. We evaluated the usability of each article that released an open-weight WAO HAR model. We faced a variety of different issues when using the open-weight models we identified in this review.

Paper	Link	Usable?	Notes and Issues
[10]	Code	✓	*Issues*: Lacks code to segment data and compute features because they provide their data as features derived from segments. *Notes*: Manually reproducing their preprocessing, segmentation, and feature extraction may be difficult but should be possible. Because the authors do not provide the raw accelerometer data, however, there is no way to verify if an implementation is computing exactly what the authors did in their original work.
[11]	Code	✓	*Same as above.*
[118]	Code	(✓)	*Issues*: Missing packages and versions, no documentation on how to use code (e.g., no `main.py` or written pipeline for preprocessing to prediction), no map for model output to WHARF activity names[26], incorrect pathing (i.e., path names refer to local directories), scripts include commented-out code, code contains no documentation. *Notes*: running these models requires a high technical proficiency. While we applaud these authors for making these resources available, we believe it would be difficult for a health researcher without substantial technical expertise to use either model independently to analyze their data.
[154]	Code	✓	N/A. This codebase is easy to run and well-documented.

Usability Rankings: ✓- this codebase is easy to use, well documented, and no errors arose when using the model for inference, (✓) - this codebase is usable; however, errors arose when using the model for inference, or the codebase is not well documented.

examples of accessible work and collaboratively reach a consensus on providing easy-to-use, well-documented supplementary materials so everyone may benefit from open source WAO HAR research.

6.1.3 How do WAO HAR Researchers Move Forward? Our review highlights a misalignment between engineers creating and researching HAR and the needs of those who desire to use these models as analytical tools. This can partially be addressed by WAO HAR researchers considering the need for replication, reproduction, and open-source algorithms within their investigation of novel methodologies.

Making research replicable, reproducible, and fully transparent is not an easy request and many fields are struggling with it (e.g., psychology [148]). Because the replication and reproduction crises are recognized as such pervasive problems, many communities have started efforts to create more replicable and reproducible research [100]. In computer science, conferences across subfields have proposed rewards to promote replication, such as the ACM VRST '25 reproduction challenge,[27] the SIGMOD Availability and Reproducibility Initiative (ARI),[28] and the Graphics Replicability Stamp initiative (GRSI).[29] These venues and external initiatives incentivize researchers to promote replication and reproduction by providing rewards and establishing competitions. Some researchers have taken the onus of promoting replication in their field on themselves. For example, Pellatt and Bock et al. [152] proposed a shared GitHub repository[30] for all deep learning in HAR, but the repository has not been updated since the original publication. This list is a small portion of replication and reproduction efforts and exemplifies how authors can be incentivized to produce replicable research.

[27] www.vrst.acm.org/vrst2025/index.php/reproduction-challenge/
[28] www.reproducibility.sigmod.org/
[29] www.replicabilitystamp.org/
[30] www.github.com/STRCSussex-UbiCompSiegen/dl_har_public

Increasing availability, comparability, and replicability of WAO HAR research will take time, effort, and collaboration; any changes to the community standards should be made and upheld collectively. Based on prior efforts to increase research replicability (e.g., WORCS [192] and the Open Source Initiative[31]) and the results of this review, we propose recommendations for changes that could be implemented to promote the availability and reproduction of future HAR work. We separate our recommendations into three different categories: those for authors/researchers, those for reviewers and publication venues, and those creating communal resources for the field. The recommendations for authors are detailed in a point-by-point checklist in Appendix A.

Recommendations for researchers: WAO HAR researchers can increase the availability, comparability, and replicability of their work by (1) including sufficient supplemental material; (2) sharing and using existing, public resources in their work; and (3) reporting and communicating their work with more precision and transparency. First, and foremost, to increase the availability and reproducibility of their work, we recommend authors make any novel algorithms (including all data and the code for preprocessing and training the resulting HAR models) discussed in their work completely open source by submitting their codebase as supplemental material.[32] Further, before submitting work for publication, the supplemental material and open-source algorithms should be verified to be usable (see Sec. 6.1.2) by an individual who did not contribute to the paper. Additionally, authors should make all models analyzed in their work open source and include code to use these models for novel inference. We recognize that releasing documented code, algorithms, and model weights may increase the time it takes to prepare and publish new research; however, we believe contributing readily available, easy-to-use WAO models would allow these models to be used by health researchers seeking to further analyze existing datasets.

Additionally, because numerous already-collected, publicly-available HAR datasets exist (see Sec. 5.1.2, [56], and [13] for such datasets), authors should defer to using them. If collecting new datasets is required (e.g., to include activities not prominent in existing datasets, or to include data from a unique population), the new datasets should be made public for other researchers to use, and authors should make efforts to test algorithms on the prior existing datasets as well. Collecting novel, private datasets similar to existing datasets not only makes the performance of the proposed models more difficult to interpret within the context of the field, but it also prevents the work from being reproduced. Together with open-source algorithms, public data allow the community to benchmark specific algorithms for tasks, thus converging on algorithmic WAO HAR solutions. Making code open source and using public data, however, is insufficient to address the replication crisis; researchers must also be diligent in reporting and communicating their work such that it can be verified and replicated by others. Venue permitting, authors must be more explicit about how their WAO HAR research in the body of the paper (see Appendix A for suggestions) so that other researchers can understand how the algorithm works as well as its benefits and limitations in comparison with other algorithms. Lastly, to ensure researchers maintain the highest quality of research reporting and transparency, we recommend all non-exploratory WAO HAR research be preregistered with the Open Science Foundation (OSF)[33] before conducting any experiments.

Although HAR researchers can vastly improve the quality of the field, they alone cannot resolve all the replication, comparison, and availability issues. As such, they must lead the effort to improve publication venue practices and broader community standards.

Recommendations for reviewers and publication venues: While publication venues and reviewers cannot do the work for researchers, they can impose criteria that would help persuade (and perhaps even require) researchers to make their work more transparent and available. Publication venues could adopt practices from other fields, such as awarding certificates/honors for papers that are exceptionally transparent and meet a minimum reporting standard; such public acknowledgment might encourage authors to submit replicable and

[31] www.opensource.org/

[32] Exceptions can and should be made for work that contains confidential or sensitive information and/or work under a patent or copyright.

[33] www.cos.io/initiatives/prereg

transparent work. Additionally, venues can require authors to submit supplemental material, ensuring an open-source algorithm and open-weight model accompany each publication. While reviewers cannot make such requirements, they can verify that authors meet the suggestions for replicable and transparent work by ensuring authors follow the guidelines outlined in Appendix A. Further, reviewers can and should read each paper explicitly for replicability (i.e., the reviewers could imagine trying to replicate or use this work, what questions would they have, and what additional information would they need); if they do not think they could replicate what the authors did, they should require the submitting authors to add necessary clarification before acceptance and publication. The review process allows authors to improve their work; reviewers should use this opportunity to ask researchers to make their work available and replicable; this process can be facilitated and expedited by publication venues setting new standards for acceptance or providing further incentives for outstanding work.

Recommendations for the community and communal resources: Beyond running experiments and publishing work, the WAO HAR community can collaborate to provide the community with overdue and necessary resources such as easily accessible dashboards or repositories highlighting public datasets, direct comparisons of different WAO HAR models' performances on specific datasets (e.g., the leaderboard in WHAR Arena [35]), and evaluation protocols and best practices. As such, we recommend the community compile a shared resource to allow researchers to submit a trained model (and accompanying open-source algorithm) that would be independently evaluated on the same benchmark datasets. This process would allow us to directly compare existing models for the same problem in a Kaggle-style[34] fashion. Direct comparison of WAO HAR algorithms would allow the HAR research community to reach a consensus on the best algorithms for general populations while affording health researchers the opportunity to make a more informed choice of WAO HAR algorithm before applying it to analyze their data. This repository should account for differences in model use cases (e.g., a personalized model should not be compared against a general one, and models capable of classifying different activities should not be directly compared) and make comparisons accordingly. Additionally, assembling a repository of online resources that link to every open-source algorithm and open-weight model, such as the repository proposed by Pellat and Bock et al. [152], would allow researchers from any discipline to easily identify, use, and adapt public WAO HAR models. Such a repository should be moderated, ensuring each contribution meets a standard of usability, and ensuring the performance of the WAO HAR models can be easily compared to other models. Guaranteeing each contribution meets a specific standard can perhaps be, at least, semi-automated, and the community should work collaboratively to establish a standard and develop such a tool. A similar tool could also ensure that authors submitting their work for publication have included all the necessary elements for replication and reproduction.

Given the scope of this review, our recommendations are framed in the context of WAO HAR literature; however, these recommendations can likely be easily adapted to promote availability and replicability across HAR research. In addition to helping our community converge on the best algorithmic solutions, open resources and available models will allow others to use HAR models as an analytic tool to understand physical activities in the wild. The significant work required to ensure replication, reproduction, and transparency in the WAO HAR literature can feel unappreciated, but it is critical to move the field forward and allow WAO HAR algorithms to be widely deployed in health and other fields; we hope authors, reviewers, publication venues, and the community can support each other in efforts to improve the utility of the work of the field.

6.2 Limitations

We conducted this review to identify and summarize the readily available, trained WAO HAR models that could be used to analyze existing large-scale, population-wide health surveillance datasets of wrist-accelerometer-only data (**RQ2**). This question guided the definition of our search terms and our exclusion criteria (specifically

[34]www.kaggle.com/.

removing WAO HAR articles that described models capable of a classification task perhaps not beneficial to health researchers (see **OOS CT** in Sec. 4.2.2)). Further, because the scope of our review was WAO HAR models, we excluded thresholding and cut-point methods because they can remove a lot of information from the raw accel signal that can be beneficial for classifying activities and are inaccurate in quantifying free-living physical activities [46, 47, 178].

Outside of our search terms and exclusion criteria, our screening process may have introduced bias that affected the articles we included in our review. Although the screening was completed by two independent authors (with a third settling disagreements or ambiguity), only one author screened each article. Thus, screening was subject to that individual author's biases, which may have resulted in some articles being excluded from the review, especially during the title and abstract screening. Before we began screening, however, we conveyed to all authors responsible for screening to err on the side of caution and send an article to the next screening stage if they were unsure; articles with non-descriptive titles or vague abstracts were sent to the full-text screening to prevent us from excluding articles prematurely. Although we had a protocol to handle uncertainty in screening, this protocol does not negate existing biases.

In addition to our review scope and screening protocols, the databases we retrieved articles from may have limited the work we reviewed because we focused on papers published primarily to a technical/engineering audience: IEEE Xplore, Science Direct, ACM DL, MDPI Sensors, and PubMed. Although methodological choices based on our primary investigation narrowed the scope and generalization of this review, we hope our results and the corresponding discussion inspire HAR researchers to employ tactics yielding more replicable, reproducible, and open-source HAR research.

6.3 Future Work

The gaps revealed by our review have paved the way for future work in WAO HAR research. Because we identified fewer available models than replicable and reproducible works, future researchers could implement and make open source the replicable and reproducible WAO HAR models for others to use. Not only will more open-source resources make it easier to compare novel solutions with existing solutions, they would also ensure that more WAO HAR models are accessible to health researchers who could use these models as analytical tools. Further, deploying these models to the public allows for a more robust validation of the models themselves (i.e., researchers could easily validate existing models on multiple independent datasets, examining limitations and the generalization of models). Additionally, because transparent and replicable WAO HAR research is currently left to individual researchers, we believe it is important for the HAR community to establish and agree upon a standardized set of guidelines for reporting the training, results, and capabilities of a novel model. Appendix A proposes some items that should be included in these guidelines. These guidelines are a first step in producing replicable work; the larger community must also ensure these guidelines are being followed by rejecting work that does not meet them and requiring codebases to be externally validated to ensure usability. Doing so would maximize the impact of WAO HAR researcher and build confidence in different algorithmic approaches so we may better understand population-wide physical activity trends. We hope this review inspires researchers to further the impact of their work by ensuring it is comparable, available, and reproducible.

7 Conclusion

Wrist-accelerometer-only (WAO) human activity recognition (HAR) models have the potential to allow us to more easily and reliably understand individual and population-wide physical activity trends. Although many WAO HAR researchers focus on preprocessing techniques or algorithm architectures, valuing improved performance through novelty, many of these publications do not evaluate their algorithms against baselines or validate them on an independent dataset. Additionally, a lack of transparency in reporting algorithms and results and a scarcity of publicly available code and open-weight models means the majority of the work we reviewed was likely neither replicable, reproducible, nor readily available. We hope this review inspires WAO HAR researchers to change the way they consider, report, and evaluate their WAO HAR algorithms, ideally yielding more open-source WAO HAR models that researchers can better understand and trust to analyze unseen datasets.

Data Availability and Protocols

We provide a copy of our protocols, detailed results from our screening protocol, our raw extracted data, additional figures, and all analyses in our supplemental material: https://anonymous.4open.science/r/wao-har-systematic-review-1663/ (anonymized for submission).

Registration

This systematic review was not preregistered.

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A Appendix A: Checklists for Authors to Improve Availability and Reproduction of their WAO HAR Work

We propose the following recommendations for *authors preparing manuscripts*. Our recommendations are broken into stages for what to include within the paper and supplemental material. Additionally, we include several items that are not necessary for available, open research, but that we believe would benefit the field.

In the *body of the paper*, authors should provide the following.Items marked with a dagger (†) should be written with sufficient detail so *anyone can replicate your work without making a judgment call.*

- ☐ **Dataset overview**: Include a section labeled “data” or “datasets” that includes all necessary components of the data: 1) collection process, 2) number of participants, 3) participant demographics, 4) what sensors were worn (e.g., brand), 5) sensor wear locations, 6) all measurements taken (including sampling rate), 7) activity names, definitions, and number, 8) how much data were collected (number of samples or minutes of each activity as an aggregate and by participant), 9) what data from the dataset were actually used (i.e., you may have used the PAMAP2 dataset which collects IMU data from three sensor locations but your work only uses accelerometer data from the right ankle). Additionally, this section should include a rationale explaining why you chose this data (especially if the data were self-collected).
- ☐ **WAO HAR contribution description**: In the research questions or contributions list, clearly state what problem in the WAO HAR community your research is investigating (e.g., is your algorithm personalized to an individual’s movements? Is the algorithm robust to location or orientation shifts? Does the algorithm perform better than the prior benchmark?).
- ☐ **Data preprocessing steps**†: In the method section, include a subsection that clearly states the following: 1) window length(s) and overlap(s),[35] 2) denoising/cleaning steps, 3) any data that was excluded and why. If applicable, also include 4) resampling methods, and 5) features extracted (by name and final feature vector length). Also include any packages/software (including versions) used during the cleaning and preprocessing steps.

[35] In the case of adaptive window size, these numbers are not necessarily reported; however, you should explain how your adaptive windowing works.

- ☐ **Algorithm details†**: In the method section, include information on the chosen algorithms, including (but not limited to): 1) architecture (as a diagram when applicable), 2) hyperparameters, and 3) training variables (e.g., number of epochs or learning rate). Also include any packages/software used during training.
- ☐ **Evaluation plan†**: In the method section, define a subsection including the word "evaluation" that describes your evaluation protocols. Include 1) evaluation strategy (e.g., single-split or LOSO) and rationale for that chosen strategy (see [31] and [101] for more information on choosing an evaluation strategy for your research question), 2) chosen metrics, and 3) what comparisons are going to be made in the results section.
- ☐ **Report precise and correct results**: In the results section, results should be reported using tables, charts, and confusion matrices. All visualizations, such as bar charts, should include the precisely computed values and error bars. Any comparisons (especially those comparing model performances from prior work) should be made as directly as possible (i.e., using the same data, with the same preprocessing steps, and the same evaluation methodologies). If your work is the first to use a specific dataset or address a novel problem, the results should be framed as a benchmark of that specific task on your dataset.
- ☐ **Supplementary material link**: Include a direct link to your supplemental material within the body of your paper. If possible, also include your supplementary material as a .zip file to be published alongside your paper in the journal/conference proceedings.

Include the following items as *supplemental material:*

- ☐ **`README.md`**: The `README.md` of your repository should provide users with an overview of the material as well as clear instructions on how to use the code/models. As such, we recommend a `README.md` have at least the following subsections: 1) repository structure (define the file organization and what users can find in each subfolder), 2) training/evaluating an algorithm (provide detailed instructions on how users can train an algorithm from your paper and evaluate the resulting model), 3) making novel inferences (provide detailed instructions on how to use a model for inference on a new dataset. Instructions should include information about what input/format the code expects data to be in and what the output format will be (e.g., if the model outputs numerical predictions, how do these numbers align to the desired activities)), 4) replicating/reproducing our results (provide instructions on what you ran to obtain your results so others can replicate it), 5) data availability (provide information on how others can obtain the data you used), 6) questions, issues, and contact (provide an easy way to reach out with questions or issues). Ideally, the `README.md` also includes information on the associated work and links to the publication.
- ☐ **`requirements.txt` (or equivalent)**: List all the packages needed to run your code, following the agreed-upon standards for the language. Include information in your `README.md` about how to install these libraries, in a new virtual environment, if applicable.
- ☐ **Source code**: Include all scripts run to obtain the results reported in your work. These scripts should include 1) data retrieval, 2) data preprocessing, 3) algorithm declaration and training, 4) model evaluation, 5) analysis scripts used to compute aggregate statistics, and 6) scripts used to produce any of your figures. Code should be clean and well-documented, according to the standard set by the language. In addition to the source code, include a document (e.g., a `.txt`) listing all arguments passed to scripts to obtain the results discussed in your paper.
- ☐ **Raw results**: Include all the raw results (e.g., a collection of `.csv` files containing predictions) obtained while running the experiments described in your paper. These files should be clearly labeled and divided into subfolders as necessary to match sections of your paper. All file organization should be documented.

- ☐ **Trained model weights**: Include final[36] weights of any models obtained/evaluated in your work. Model weights may be large; we recommend compressing them and/or sharing them through a third party (e.g., Google Drive or Hugging Face).
- ☐ **Demonstration script**: Include a script that loads model weights, preprocesses some data, uses the loaded model for inference, and saves the predictions to a specified location. Ideally, the script should load publicly available data and include the location where users can download the data. The script should include comments noting which lines users of the code are likely to change (e.g., data, desired model to use, file path to save results).
- ☐ **Additional writing**: In the case that a venue has a word/page limit, rhetoric that should be included in a manuscript (see above), but was cut for space, should be reported in a supplemental document or an appendix.

Additional miscellaneous items to promote availability and reproduction:

- ☐ **Verify supplemental materials are usable**: Have an external party (not a co-author) review your supplemental material. Ideally, they should be able to run your scripts and generate results or novel predictions without your help. If they have questions, you should improve your supplemental material.
- ☐ **Preregister your work**: Preregister your work (including research questions, analysis plan, and the dataset) before running experiments.[37] The link to the preregistration should be included in your paper or the supplemental material. If your work is not preregistered, declare so.

[36]Because it is unreasonable to supply weights from all models obtained during LOSO CV (or a similar evaluation), we use the term final to refer to the model obtained by training the algorithm on *all* available data in the dataset.
[37]We recommend using the Center for Open Science (www.cos.io/initiatives/prereg).